

1. Introduction & Foundations

- Introduction-What We will Learn In This Course
- What is Generative AI?
- What is Agentic AI?
- Generative AI vs Agentic AI

2. Fundamentals of GenAI and LLM's

- Basics of Generative AI block chapter
- What are LLMs?

Key Concepts

- Tokenization
- Embeddings
- Vocabulary
- Attention

Transformer Architecture

- Self-Attention Mechanism
- Multi-head attention
- Cross Attention
- Encoder – Decoder

3. Working With LLM's

- Open Source Vs Closed Source LLMs
- HuggingFace Ecosystem
 - Model Loading
 - Model Parameters
 - Model Weight Formats (.pth, safetensors, onnx)
 - Model Size calculation and Licence
- Multi-modal LLMs: Text, Audio (ASR, TTS), Image, Video

4. Prompt Engineering & Control

Prompts and Context

- Max Sequence Length Vs Max Output Tokens
- Task-specific Prompts

Sampling Parameters (Temperature, Top-k, Top-p, etc.)

Prompting Techniques

- Zero-Shot Learning
- Chain of Thought (CoT)
- React
- Guardrails

Using Chat Completion APIs

- a. OpenAI (ChatGPT)
- b. Google Gemini
- c. Anthropic Claude

5. Retrieval-Augmented Generations-RAG

- o What is RAG?

LLM Hallucination: Causes and Mitigation

When to Use RAG

Components of RAG:

- a. Embeddings
- b. Vector Databases: Chroma, Pinecone, FAISS, Milvus
- c. Chunking Strategies
- d. Conversational RAG
- e. Embedding Spaces: Semantic Similarity & Cosine Distance
- f. Answer Grading / Response Evaluation (BLEU, ROUGE, GPT-based)

6. Advanced RAG Patterns

- a. Corrective RAG (CRAG)
- b. Self-RAG with Reflection
- c. Graph RAG
- d. Hybrid RAG(Semantic + Keyword)

7. Graph-Based Knowledge & Retrieval

- a. Graph Fundamentals – Nodes, Edges
- b. Ontology Design
- c. GraphDBs:
 - i. Advantages of GraphDB
 - ii. Neo4j: Community vs Enterprise vs Cloud

8. Lang Chain Framework

- o Introduction to LangChain
- o Chains, Prompts, and Templates
- o Memory Systems and Conversation Flow
 - Memory Types: Short-term, Long-term, Episodic o Persistence Strategies
 - Persistence Strategies
- o Basic Document Loaders and Text Splitters
- o Tool Calling and Structured Output
- o Output Parsers
- o Building RAG Pipelines with LangChain

9. Agentic AI Principles

- o What is Agentic AI?
- o AI Agents vs Agentic AI
- o **Agentic AI Frameworks Overview**
 - No-code vs Code-first
 - N8n vs LangGraph vs Airflow

- **Design Principles**
 - Goal, Planner, Orchestrator
 - Copilot vs Autopilot
 - Human in loop

10. Agentic AI Frameworks

- a. CrewAI, LangGraph, AutoGen

11. LangGraph - Building Agent Workflows

- Agentic AI using LangGraph
- LangChain vs LangGraph
- LangGraph Components
- **Workflow Types:**
 - Sequential
 - Parallel
 - Iterative
 - Conditional
- **Memory & State Management**
 - Persistence Strategies
 - Time Travel in LangGraph
- **Observability with LangSmith**
 - What is Observability?

12. Model Context Protocol

- a. MCP Fundamentals and Architecture
- b. MCP Server and Client Implementation
- c. Tool Integration through MCP
- d. MCP vs Traditional Integration Patterns

13. Agent -to-Agent Communication (A2A)

- A2A Communication Fundamentals
- **Orchestration Patterns:**
 - Manager-Worker
 - Peer-to-Peer

14. Fine-Tuning and Quantization

- a. **When to Use Fine-tuning vs Prompt Engineering**
- b. **Parameter-Efficient Fine-Tuning (PEFT)**
 - i. LoRA
 - ii. QLoRA
- c. **Quantization Techniques**
 - i. Intro to Quantization
 - ii. Asymmetric vs Symmetric
 - iii. Post-training Quantization vs Quantization-Aware

15. Model Serving & Deployment

Model Serving Frameworks:

- a. vLLM
- b. TensorRT-LLM

Python Syllabus (For GenAI & AI Development)

1. Python Basics

Core Concepts

- What is Python & Installation
- Variables & Data Types
 - int, float, string, boolean
- Type Casting

Operators

- Arithmetic
- Comparison
- Logical

Control Flow

- if, elif, else
- Nested conditions

Loops

- for loop
- while loop
- break, continue, pass

2. Data Structures (Very Important)

◆ Lists

- Create, access, modify
- List methods

◆ Tuples

- Immutable data
- Use cases

◆ Sets

- Unique values
- Set operations

◆ Dictionaries (Most Important)

- Key-value pairs
- Access, update
- Iteration

3. Functions

- Defining functions
- Parameters & return values

- Default arguments
- Lambda functions

4. File Handling

- Read files (read, readline)
- Write files (write, append)
- Working with text files

5. Exception Handling

- try, except
- finally
- Custom exceptions

6. Object-Oriented Programming (OOP)

- Classes & Objects
- Constructors (__init__)
- Inheritance
- Encapsulation
- Polymorphism

7. Modules & Packages

- Importing modules
- Creating custom modules
- Using built-in modules

8. Working with APIs (Very Important for GenAI)

- What is REST API
- HTTP methods (GET, POST)
- Using requests library

Practice:

- Call API
- Parse JSON response

9. JSON Handling

- Load JSON (json.load)
- Convert to dictionary
- Serialize data

10. Data Handling for AI

◆ NumPy (Basics)

- Arrays
- Basic operations

◆ Pandas (Basics)

- DataFrames
- Reading CSV files
- Filtering data

11. Python for Text Processing (Important for LLMs)

- String operations
- Token splitting
- Cleaning text

- Regular expressions (basic)

12. Python for GenAI Integration

- Calling LLM APIs
- Handling responses
- Prompt formatting

13. Async Programming (Optional but Useful)

- Async/await
- Handling multiple API calls

Capstone Projects:-

- AI-Powered Developer Assistant
- AI Real Estate Recommendation System
- Multi-Agent Customer Support System
- AI Resume Analyzer & Interview Coach
- DevOps AI Assistant
- AI Log Analyzer
- AI Product Recommendation System
- AI Code Review Agent
- AI Knowledge Base Chatbot
- End-to-End GenAI Platform

Course Highlights:-

1. LMS Portal Access (24/7 Learning)
2. Daily Practice Exams & Assessments
3. Weekend Workshops (Hands-on Sessions)
4. Real-Time Projects (Industry-Based)
5. Internship Opportunity
6. Placement Assistance (100% Support)
7. Expert Trainers with Industry Experience
8. Doubt Clarification Sessions
9. Live Coding & Practical Sessions
10. Resume Building & Interview Preparation
11. Mock Interviews & HR Training
12. Certification on Course Completion
13. GitHub & Portfolio Development
14. Hackathons & Idea Pitching Sessions
15. Career Guidance & Mentorship